

Online Appendix to “Process Efficiency as a Barrier to Innovation”

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This Online Appendix accompanies the paper. Section, equation, table, figure, proposition and footnote numbers that carry no appendix letter refer to the main text. Appendix A documents the allocation of subsidies and patents across firm size; B collects the production-side algebra; C the proofs; D the dynamic treatment of process efficiency; E the breakthrough-patent measure; F the reduced-form robustness checks; G the standard-error construction; H supplementary estimation results, including the restricted model, the model-implied crowding-out coefficient and the calibrated economy; I the computational algorithm; J the robustness of the quantitative results; and K the firm-type heterogeneity extension.

Appendix A. The Allocation of Subsidies and Patents Across Firm Size

This appendix documents the allocation of innovation subsidies and patent output across the firm size distribution in China, referenced in Sections 3 and 5.

Firms are grouped using the ten employment-size bins defined by the Chinese National Bureau of Statistics in its Economic Census size classification. Bin-level comparisons alone would be confounded by cross-industry composition: certain sectors are systematically larger and more subsidy-intensive than others. I therefore compute all comparisons within 4-digit industry $\times$ year cells, considering only cells with at least 10 firms, yielding a sample of 1,889 industry-years. In this version I reweight firms so that each of the ten employment-size bins carries equal weight, which compares size classes on the footing of the average firm rather than economy-wide totals; the subsidy-yield panels are unaffected by the reweighting. The two patterns documented below are robust, in results not reported here, to re-weighting the stratified CTS sample to match the firm-size bin shares of the Chinese National Economic Census and to restricting the sample window to 2016–2020.

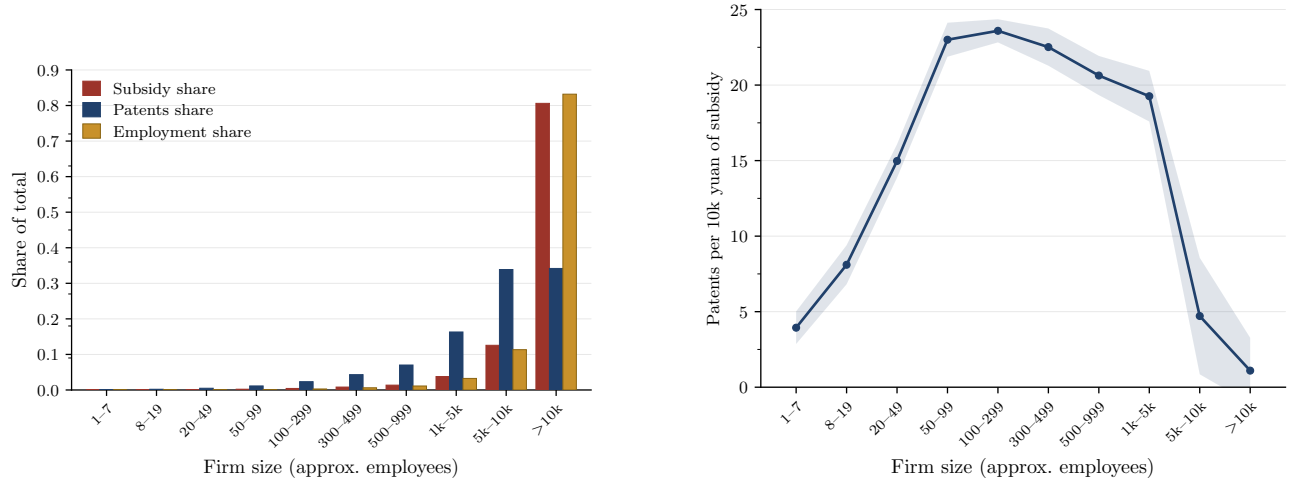
A.1 Subsidy and patent shares

For each 4-digit industry $\times$ year cell with at least 10 firms, I compute the share of subsidies, patents, and employment accruing to each of the ten NBS employment-size bins, reweighting firms so that every

bin carries equal weight. Panel (a) of Figure A.1 shows the result. With the bins weighted equally, the over-10,000-employee bin accounts for 81% of subsidies and 83% of employment but only 34% of patents. Across the mid-size range patent shares exceed subsidy shares, with the 1,000–5,000-employee bin accounting for 16% of patents on 4% of subsidies. The sharpest break is between the 5,000–10,000-employee bin and the bin above 10,000 employees: the subsidy share jumps from 13% to 81% and the employment share from 11% to 83%, while the patent share stays essentially flat near 34%. Subsidies and employment thus track one another across firm size and diverge sharply from patents.

A.2 Patents per subsidy yuan

I define *subsidy yield* as the number of patents produced per 10,000 yuan of subsidy received, computed among firms receiving positive subsidies within each 4-digit industry $\times$ year cell.¹ Panel (b) of Figure A.1 plots subsidy yield against firm size. The relationship is hump-shaped: yield rises from roughly 4 patents per 10,000 yuan at the smallest firms to a peak near 24 at 100–200 employees, remains high through the mid-size range, and falls toward zero above 3,000 employees. The gap between the peak and the right tail lies outside the 95% confidence bands at each end.



(a) Distribution of subsidies, patents, and employment across firm size.

(b) Patents per 10,000 yuan of subsidy by firm size.

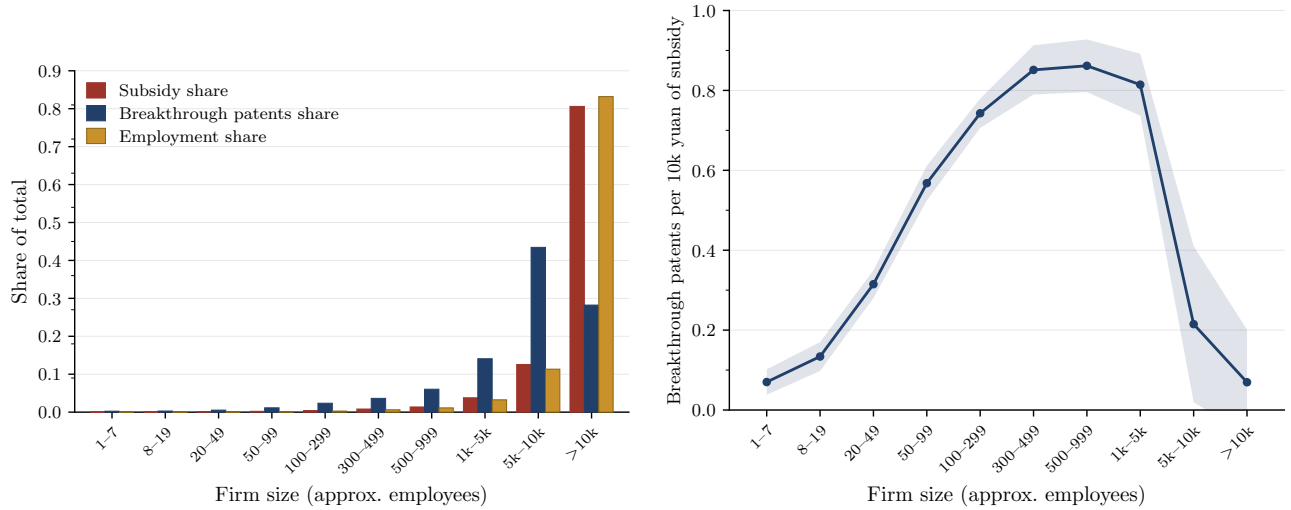
Figure A.1: Subsidy allocation and yield across firm size

Notes: Panel (a) shows the shares of subsidies, patents, and employment accruing to each NBS employment-size bin when firms are reweighted so that every bin carries equal weight. Panel (b) shows patents per 10,000 yuan of subsidy among recipients, with the 95% confidence band computed from between-cell variation; the yield profile is invariant to the reweighting. Yields are winsorized at the 99th percentile. Bin-level statistics require at least three subsidized firms per cell and five contributing cells per bin.

¹Yield is defined only for recipients. If selection into subsidies correlates with unobserved innovation capacity, so that small subsidized firms are more capable than the typical small firm, the hump could partly reflect selection rather than the marginal productivity of a subsidy yuan. The reduced-form analysis in Section 3.2 addresses this by using the full firm population and exploiting within-industry variation in subsidy concentration.

A.3 Robustness to strategic patenting

One concern with these patterns is that they reflect strategic patenting rather than genuine innovation. Chinese firms have well-documented incentives to file low-quality patents to qualify for High-Technology Enterprise designation, tax preferences, and procurement eligibility. If small and mid-size firms file strategically at higher rates than large firms, the patterns above could be artifacts of the filing margin. To address this, I reconstruct both panels of Figure A.1 using only breakthrough patents, defined as those in the top 10% of year-demeaned log inventiveness by the Kelly et al. (2021) measure, adapted to Chinese patent text (Appendix E provides construction details). Figure A.2 shows that both patterns are unchanged. The over-10,000-employee bin accounts for 81% of subsidies but only 28% of breakthrough patents, with breakthrough patenting concentrated in the 1,000–10,000-employee bins (14% and 43%, peaking at 5,000–10,000). Breakthrough patents per subsidy yuan display the same hump-shaped profile: yield is highest for mid-sized firms with 200–2,500 employees and declines toward zero above 5,000 employees. The mismatch between the allocation of subsidies and the location of patent output is therefore not driven by low-quality filings.



(a) Distribution of subsidies, breakthrough patents, and employment.

(b) Breakthrough patents per 10,000 yuan of subsidy.

Figure A.2: Subsidy allocation and breakthrough patent yields across firm size

Notes: Breakthrough patents are the top decile of year-demeaned log inventiveness (Appendix E). In panel (a), each bar shows the share of subsidies, breakthrough patents, or employment accounted for by firms in that size bin when the bins are reweighted to equal firm weight. Panel (b) reports yields among subsidy recipients, winsorized at the 99th percentile before aggregation, with the shaded 95% confidence band. All statistics are computed within 4-digit industry $\times$ year cells.

These patterns are partial equilibrium: they show that subsidies and patents do not co-locate across the firm size distribution, but they cannot establish whether subsidies to large firms suppress patenting elsewhere in the economy. Section 3.2 provides that evidence.

Appendix B. Production-Side Algebra

This appendix derives the profit coefficient $\tilde{p}$ used in Section 2.2. Given prices $\{p_j\}$, the final-good producer chooses intermediate quantities to solve

$$\max_{\{k_j\}} L^\beta \int_{j \in \mathcal{N}} z_j^\beta k_j^{1-\beta} dj - \int_{j \in \mathcal{N}} p_j k_j dj,$$

with first-order condition

$$k_j = (1 - \beta)^{1/\beta} L z_j p_j^{-1/\beta}.$$

Demand has constant elasticity $1/\beta$ and every line has marginal cost $w^u/\bar{q}$, so the leader charges $p_j = w^u/[(1 - \beta)\bar{q}]$ on every active line. Substituting,

$$k_j = (1 - \beta)^{2/\beta} L z_j \bar{q}^{1/\beta} (w^u)^{-1/\beta}, \quad \Pi_j = \frac{\beta}{1 - \beta} \frac{w^u}{\bar{q}} k_j = \beta (1 - \beta)^{\frac{2}{\beta}-1} L z_j \bar{q}^{\frac{1}{\beta}-1} (w^u)^{1-\frac{1}{\beta}}.$$

Since $z_j^\beta k_j^{1-\beta} \propto z_j$, aggregate output is $Y = (1 - \beta)^{2(1-\beta)/\beta} L \bar{q}^{(1-\beta)/\beta} (w^u)^{-(1-\beta)/\beta} Z$, where $Z \equiv \int_{j \in \mathcal{N}} z_j dj$. The assembly first-order condition $w^u = \beta Y/L$ pins down the wage,

$$w^u = \beta^\beta (1 - \beta)^{2(1-\beta)} \bar{q}^{1-\beta} Z^\beta,$$

and substituting back, with the powers of $(1 - \beta)$ collecting to $\frac{2-2(1-\beta)^2}{\beta} - 1 = 3 - 2\beta$ and those of $\bar{q}$ to $\frac{1-(1-\beta)^2}{\beta} - 1 = 1 - \beta$,

$$\Pi_j = \beta^\beta (1 - \beta)^{3-2\beta} L z_j \bar{q}^{1-\beta} Z^{\beta-1}. \quad (\text{B.1})$$

Equation (B.1) is linear in $z_j = q_j(1 + \psi_f)$, so for an arbitrary quality distribution the normalized line profit is

$$\frac{\Pi_j}{\bar{q}} = \tilde{p} \frac{q_j}{\bar{q}} (1 + \psi_f), \quad \tilde{p} = \beta^\beta (1 - \beta)^{3-2\beta} L (Z/\bar{q})^{\beta-1}.$$

Summing over the firm's portfolio gives operating profit $\tilde{p}(1 + \psi_f) Q_f$, the expression used in Section 2.2. On the symmetric balanced growth path, $q_j = \bar{q}$ on active lines, so $Z = \bar{q} \Phi_A \bar{\Psi}$, since active lines have measure Φ_A and the line shares $s(n)$ average $1 + \psi^*(n)$ to $\bar{\Psi}$. Substituting delivers the closed form

$$\tilde{p} = \beta^\beta (1 - \beta)^{3-2\beta} L (\Phi_A \bar{\Psi})^{\beta-1}, \quad (\text{B.2})$$

which firms take as given in Section 2.3; since $\beta < 1$, higher average process efficiency lowers $\tilde{p}$. Also $Q_f = n_f$, so a firm with n lines and process efficiency ψ_f earns $\tilde{p}(1 + \psi_f) n$ before R&D costs.

Two features are used elsewhere. Output is $Y = \beta^{\beta-1} (1 - \beta)^{2(1-\beta)} L \bar{q} (\Phi_A \bar{\Psi})^\beta$, so Y and $\bar{q}$ grow at the same rate on the balanced growth path, as (19) assumes. Unskilled labor used in intermediate production is $\int_{\mathcal{N}} k_j / \bar{q} dj = (1 - \beta)^2 L / \beta$, a constant multiple of assembly labor, as claimed in footnote 2.

Appendix C. Proofs

This appendix collects proofs of the propositions stated in the main text, together with a derivation of the three growth channels discussed in Section 2.5. The proof of Proposition 1 rests on Lemma 1, which holds for a defender of arbitrary aggregate quality facing any fixed challenger intensities; the proofs of Propositions 2 and 3 are stated on the symmetric balanced growth path, where $Q = n$ and the static objects of Section 2.2 are evaluated at $Q = n$.

Proof of Proposition 1 (Process efficiency insulates large incumbents).

The monotonicity holds for any fixed set of challengers, which I record as a lemma.

Lemma 1. *Let a defender with aggregate quality Q face challengers i with attempt intensities $a_i \geq 0$ and process efficiencies $\psi_i \geq 0$, so that its per-line displacement rate is*

$$T(Q) = \sum_i a_i \Lambda((1 + \lambda)(1 + \psi_i) - (1 + \psi^*(Q))).$$

Then T is decreasing in Q . In the step-function limit $\kappa \rightarrow \infty$, a challenger with $\psi_i = 0$ contributes zero whenever $\psi^(Q) > \lambda$.*

Proof of the lemma. The defender enters T only through $\psi^*(Q)$, which is strictly increasing in Q by (5). Each term's argument $(1 + \lambda)(1 + \psi_i) - (1 + \psi^*(Q))$ is therefore decreasing in Q , and Λ is increasing, so each term is decreasing in Q ; a sum with nonnegative weights inherits the monotonicity. For $\psi_i = 0$ the argument is $\lambda - \psi^*(Q)$, and as $\kappa \rightarrow \infty$, $\Lambda(\lambda - \psi^*(Q)) \rightarrow \mathbf{1}\{\lambda > \psi^*(Q)\}$ pointwise away from the knife-edge $\psi^*(Q) = \lambda$, and the limit is zero whenever $\psi^*(Q) > \lambda$. ■

Evaluating the lemma at the equilibrium challenger pool—entrants with intensity X_e and $\psi = 0$, and size- n'' incumbents with intensities $F\mu(n'')X^*(n'')$ and efficiencies $\psi^*(n'')$ —gives the displacement rate of equation (15), so $\tau(Q)$ is decreasing in Q . For $Q > \bar{Q}_e$, equation (13) gives $\psi^*(Q) > \lambda$, so by the lemma no entrant can displace the firm in the step-function limit; under the logistic smoother the entry term is positive but small.² Among incumbent challengers, displacement requires $(1 + \lambda)(1 + \psi^*(n'')) > 1 + \psi^*(Q)$, and because the required $\psi^*(n'')$ rises with the defender's Q , the set of qualifying challengers shrinks as Q grows. ■

Proof of Proposition 2 (Larger firms devote more R&D labor to process efficiency).

From (6), process R&D labor demand is $h_\psi(n) = c_\psi[\psi^*(n)]^\omega \propto n^\sigma$ exactly, with $\sigma = \omega/(\omega - 1)$. Product R&D labor has no closed form, so I characterize it by seeking a solution of (10) that is asymptotically a power law, $\hat{V}(n) \sim V_0 n^b$ for some $b > 0$ and $V_0 > 0$, and then verify which exponent is consistent.

²At $Q = \bar{Q}_e$ the entrant's advantage is zero and $\Lambda(0) = 1/2$, so the threshold is sharp only in the limit. At the estimated parameters the entrant success probability falls to 0.11 at $Q = 1.5 \bar{Q}_e$ and to 0.02 at $Q = 2 \bar{Q}_e$.

Under the ansatz $\Delta\hat{V}(n) \sim b V_0 n^{b-1}$. Since $B(n)$ is bounded between Φ_I and one, (11) gives $X^*(n) = O(n^{(1-\gamma)(b-1)/\gamma})$ and hence $h_X(n) = (X^*(n)/\theta)^{1/(1-\gamma)} = O(n^{(b-1)/\gamma})$. Collecting orders on the right-hand side of (10), the flow payoff is $O(n^{\max\{\sigma, (b-1)/\gamma\}})$, the displacement term $n \tau(n) \Delta\hat{V}(n)$ is $O(n^b)$ with $\tau(n)$ bounded and decreasing by Proposition 1, and the innovation term $X^*(n)B(n)\Delta\hat{V}(n)$ is of lower order whenever $(1-\gamma)(b-1)/\gamma < 1$. Consistency therefore requires

$$b = \max \left\{ \sigma, \frac{b-1}{\gamma} \right\},$$

which has two roots: $b = \sigma$ and $b = 1/(1-\gamma)$.

Assumption 1 is equivalent to $\sigma < 1/(1-\gamma)$, since $\sigma = \omega/(\omega-1)$ is decreasing in ω and equals $1/(1-\gamma)$ at $\omega = 1/\gamma$. Both roots are therefore admissible under the assumption, and the second is the larger. I rule it out as follows. At $b = 1/(1-\gamma)$ the first-order condition delivers $X^*(n) = O(n)$ and $h_X(n) = O(n^{1/(1-\gamma)})$, so the flow payoff $\pi(n) - w_s h_X(n) = O(n^\sigma) - O(n^{1/(1-\gamma)})$ diverges to $-\infty$. Firm value is then positive and growing while current payoffs are unboundedly negative, sustained only by the anticipation of further growth. Imposing the transversality condition $\lim_{T \rightarrow \infty} e^{-\hat{r}T} \mathbb{E}[\hat{V}(n_T)] = 0$ eliminates this branch. It is also not the branch the model is solved on: the policy iteration of Appendix I starts from $\hat{V} \equiv 0$ and converges to the first root, with $d \log \hat{V} / d \log n$ approaching σ at the estimated parameters.

On the surviving branch $b = \sigma$, so $\Delta\hat{V}(n) = O(n^{\sigma-1})$,

$$X^*(n) = O(n^\alpha), \quad \alpha = \frac{(1-\gamma)(\sigma-1)}{\gamma} = \frac{1-\gamma}{\gamma(\omega-1)},$$

and $h_X(n) = O(n^{\alpha/(1-\gamma)}) = O(n^{1/[\gamma(\omega-1)]})$. Assumption 1 also secures the lower-order condition used above, since $\sigma < 1/(1-\gamma)$ implies $(1-\gamma)(\sigma-1) < \gamma$ and hence $\alpha < 1$.³

Comparing the two exponents,

$$\sigma > \frac{1}{\gamma(\omega-1)} \iff \frac{\omega}{\omega-1} > \frac{1}{\gamma(\omega-1)} \iff \omega > \frac{1}{\gamma},$$

so Assumption 1 is exactly the condition under which h_ψ grows at a strictly faster polynomial rate than h_X . Then $h_X(n)/h_\psi(n) \rightarrow 0$ and $p(n) = h_\psi(n)/[h_\psi(n) + h_X(n)] \rightarrow 1$, so process efficiency absorbs the whole of a large firm's R&D budget in the limit. ■

The limit does not by itself establish monotonicity, since the ratio h_X/h_ψ carries lower-order terms that need not vanish monotonically. At the estimated parameters $p(n)$ is increasing across the full computed range (Figure H.2a), and the counterfactuals in Section 5 rely on that computed profile rather than on the asymptotic result.

³At the estimated parameters $\alpha = 0.44$ and the exponent on h_X is 0.88, against $\sigma = 1.44$ for h_ψ .

Proof of Proposition 3 (Subsidies to large firms fund process efficiency).

Consider a marginal subsidy to a firm of size n , lowering its effective skilled wage to $(1 - \sigma_s)w_s$. The firm is atomistic, so it takes the aggregate wage w_s , its success probability $B(n)$, and its continuation value $\hat{V}(\cdot)$ as given, and its response is a movement along its own best-response schedule.

Process efficiency solves (5) at the subsidized wage, giving $\psi^*(n; \sigma_s) = (1 - \sigma_s)^{-1/(\omega-1)}\psi^*(n)$ and hence $h_\psi(n; \sigma_s) = c_\psi[\psi^*(n; \sigma_s)]^\omega = (1 - \sigma_s)^{-\sigma}h_\psi(n)$. Product R&D labor solves (11); with $B(n)$ and $\hat{V}(\cdot)$ fixed, $h_X(n; \sigma_s) = (1 - \sigma_s)^{-1/\gamma}h_X(n)$. Differentiating each at $\sigma_s = 0$,

$$\frac{\partial h_\psi}{\partial \sigma_s} = \sigma h_\psi(n), \quad \frac{\partial h_X}{\partial \sigma_s} = \frac{1}{\gamma} h_X(n),$$

so the share of the additional R&D labor going to process efficiency is

$$q(n) = \frac{\partial h_\psi / \partial \sigma_s}{\partial h_\psi / \partial \sigma_s + \partial h_X / \partial \sigma_s} = \frac{\gamma \sigma p(n)}{\gamma \sigma p(n) + (1 - p(n))},$$

where $p(n) = h_\psi(n)/[h_\psi(n) + h_X(n)]$. Differentiating, $q'(p) = \gamma \sigma / [\gamma \sigma p + 1 - p]^2 > 0$, so $q(n)$ is strictly increasing in $p(n)$.

Proposition 2 gives $p(n) \rightarrow 1$ under Assumption 1, and q is strictly increasing in p , so $q(n) \rightarrow 1$: in the limit the entire marginal R&D dollar at a large subsidized firm goes to process efficiency. Over the computed size range $q(n)$ inherits the monotonicity of $p(n)$. ■

Derivation of the growth channels in Section 2.5

Channel (a): Direct innovation (positive). The subsidy lowers the effective wage at subsidized firms, raising $X^*(n)$ through (11), both because the marginal cost of R&D falls and because $\Delta \hat{V}(n)$ rises with the higher continuation value. The aggregate innovation rate $\bar{\tau}$ increases, raising g in (19).

Channel (b): Blocking (negative). Under the subsidy $\psi^*(n; \sigma_s) > \psi^*(n)$ at subsidized firms. From (12), the advantage $A_{n'',n}$ falls for any challenger n'' innovating against a subsidized incumbent, so $\Lambda(A_{n'',n})$ falls for those attempts. Subsidized firms hold a positive share of product lines, so the aggregate success probability $\bar{B}$ falls and g with it.

Channel (c): Crowding out (negative). By Proposition 3, the increase in labor demand is concentrated disproportionately in process R&D. Aggregate skilled labor demand rises, and with supply fixed at L^S the wage w_s must rise to restore clearing (18). The higher wage reduces $X^*(n)$ at unsubsidized firms and reduces entry X_e , lowering $\bar{\tau}$.

Appendix D. Static vs. Dynamic Choice of Process Efficiency

The optimal process efficiency (5) solves (4), which treats ψ as affecting only current-period profit. Throughout this appendix I work on the symmetric balanced growth path, where $Q = n$, and write the static objects as functions of n . Here I allow ψ to affect continuation values as well, and show that

the static formulation understates $\psi^*(n)$, by more at larger firms over the empirically relevant range. Since the omission attenuates the blocking mechanism, the quantitative results in the main text are conservative.⁴

D.1 Dynamic value function

Let $\hat{V}(n; \psi)$ denote the value, normalized by $\bar{q}$, of a size- n firm maintaining process efficiency ψ . Allowing ψ to affect both attack and defense generalizes (10) to

$$\hat{r} \hat{V}(n; \psi) = \tilde{p}(1 + \psi)n - w_s c_\psi \psi^\omega - w_s h_X(n) + X^*(n) B(n; \psi) \Delta V_+(n) - n \tau(n; \psi) \Delta V_-(n), \quad (\text{D.1})$$

where $\Delta V_+(n) \equiv \hat{V}(n+1) - \hat{V}(n)$ and $\Delta V_-(n) \equiv \hat{V}(n) - \hat{V}(n-1)$, with continuation values evaluated at the re-optimized $\psi^*(\cdot)$ of the corresponding size. Two objects in (D.1) depend on the firm's own ψ . Its success rate, from (14),

$$B(n; \psi) = \Phi_I + \Phi_A \sum_{n'} s(n') \Lambda((1 + \lambda)(1 + \psi) - (1 + \psi^*(n'))),$$

is increasing in ψ , since a higher own ψ raises effective productivity on every target. Its line-loss rate,

$$\tau(n; \psi) = X_e \Lambda(\lambda - \psi) + \sum_{n''} F \mu(n'') X^*(n'') \Lambda((1 + \lambda)(1 + \psi^*(n'')) - (1 + \psi)),$$

is decreasing in ψ , since a higher own ψ raises the bar for every challenger.

Both the argument below and the first-order condition (11) in the main text use the monotonicity of firm value, which I record here.

Lemma 2. *$\hat{V}$ is strictly increasing in n , with $\hat{V}(0) = 0$.*

Proof. A firm that loses its last line holds nothing and never innovates again, so $\hat{V}(0) = 0$. For the monotonicity, let a firm of size $n+1$ follow the product-R&D policy of a size- n firm. The value from this policy is a lower bound on $\hat{V}(n+1)$. Subject both firms to the same realizations of innovation and displacement shocks, matching the smaller firm's lines with n of the larger firm's. The two firms then attempt at the same rate, while the larger succeeds with weakly higher probability (B is increasing in size by (14)) and loses each line at a weakly lower hazard (τ is decreasing in size by Proposition 1). The larger firm therefore holds at least one line more than the smaller until the extra line is lost, and the two portfolios coincide afterward. Since flow profit π is strictly increasing, the larger firm earns strictly more while ahead and the same thereafter, so $\hat{V}(n+1) > \hat{V}(n)$. ■

⁴Process efficiency is still re-optimized at every instant, so ψ never becomes a state variable. A formulation in which today's ψ persists and constrains tomorrow's is a different problem, which I do not solve here.

D.2 Dynamic first-order condition

Differentiate (D.1) with respect to ψ . The continuation values are evaluated at the re-optimized ψ^* of the corresponding sizes and so do not vary with the firm's current ψ , and the envelope theorem removes the induced adjustment of X^* . At the optimum $\partial \hat{V}(n; \psi)/\partial \psi = 0$, and the first-order condition for ψ is

$$\underbrace{\omega w_s c_\psi \psi^{\omega-1}}_{L(\psi)} = \underbrace{\tilde{p}n + \underbrace{X^*(n) \frac{\partial B}{\partial \psi} \Delta V_+(n)}_{\text{Attack term } \mathcal{A}(n; \psi)} + \underbrace{n \left| \frac{\partial \tau}{\partial \psi} \right| \Delta V_-(n)}_{\text{Defense term } \mathcal{D}(n; \psi)}}_{R(n; \psi)}. \quad (\text{D.2})$$

Setting $\mathcal{A} = \mathcal{D} = 0$ recovers (5).

Proposition D.1 (Static formulation is a lower bound). *Equation (D.2) has at least one solution, and every solution satisfies $\psi_{\text{dyn}}^*(n) > \psi_{\text{stat}}^*(n)$ for every $n \geq 1$.*

Proof. For existence, $L(0) = 0$ while $R(n; 0) \geq \tilde{p}n > 0$. The logistic slope is bounded, $\Lambda' \leq \kappa/4$, so $\partial B/\partial \psi \leq (1 + \lambda)\kappa/4$ and $|\partial \tau/\partial \psi| \leq (X_e + F \sum_{n''} \mu(n'') X^*(n''))\kappa/4$; with $\Delta V_\pm(n)$ finite, $R(n; \cdot)$ is bounded above. Since L is continuous and unbounded and $R(n; \cdot)$ is continuous and bounded, $L - R$ changes sign, and a root exists by the intermediate value theorem.

For the bound, both correction terms are strictly positive at every ψ . For the attack term, $X^*(n) > 0$ for $n \geq 1$ by (11), $\partial B/\partial \psi > 0$, and $\Delta V_+(n) > 0$ by Lemma 2. For the defense term, $n \geq 1$, $|\partial \tau/\partial \psi| > 0$, and $\Delta V_-(n) > 0$ by the same lemma. Hence $R(n; \psi) > \tilde{p}n$ for every ψ . Let ψ_{dyn} be any root. Then

$$L(\psi_{\text{dyn}}) = R(n; \psi_{\text{dyn}}) > \tilde{p}n = L(\psi_{\text{stat}}),$$

where the last equality is the static first-order condition. For $\omega > 1$, L is strictly increasing on $\psi > 0$ and therefore injective, so $\psi_{\text{dyn}} > \psi_{\text{stat}}$. $\blacksquare$

The argument places no monotonicity requirement on $R(n; \cdot)$, which is not monotone because Λ' rises and then falls in ψ . Multiple roots are therefore possible, and every one of them exceeds the static solution.

The gap widens with firm size over the range relevant for the quantitative exercise. The defense term carries a direct factor of n , and $\Delta V_-(n) = O(n^{\sigma-1})$ with $\sigma > 1$ by the asymptotics in the proof of Proposition 2, so it grows with size; larger firms also lose lines more slowly (Proposition 1). The attack term is positive throughout, with total product R&D $X^*(n)$ rising in n even as per-line intensity falls. Both corrections scale with Λ' , which decays once a firm's advantage places every contest deep in the tail of the smoother, so they fade for a fully insulated firm. Note that the calibrated economy is far from that region, with a blocking probability of 0.13 at the largest firms (Section 4.2), and $\mathcal{A}(n) + \mathcal{D}(n)$ at the estimated parameters rises over the observed size range.

The attack term also applies to prospective entrants, who would acquire $\psi > 0$ to raise B_e rather than attacking with $\psi = 0$ as in the baseline. This works against blocking. I therefore read what

follows as applying to the incumbent side, where both corrections operate, and treat the net effect on entry as ambiguous.

D.3 Implications for the paper’s results

1. **The size gradient in process-efficiency R&D (Proposition 2) is understated.** Process R&D labor demand $h_\psi(n) = c_\psi[\psi^*(n)]^\omega$ is monotone in $\psi^*(n)$, which is larger under the dynamic formulation and rises more steeply over the relevant size range, so the fraction $p(n)$ of R&D labor devoted to process efficiency does too.
2. **The blocking externality (Proposition 1) is understated.** The advantage $A_{n'',n} = (1 + \lambda)(1 + \psi^*(n'')) - (1 + \psi^*(n))$ falls in the defender’s $\psi^*(n)$, so a larger $\psi^*(n)$ at large firms raises the blocking probability $1 - \Lambda(A_{n'',n})$ facing every challenger n'' .
3. **The welfare gains from reallocation are lower bounds.** The gains in Section 5 operate through both channels, and both are attenuated under the static formulation.

Appendix E. Breakthrough Patent Construction

This appendix constructs the breakthrough-patent measure used in Appendix A and Section 3.2. The measure is the text-based inventiveness score of Kelly et al. (2021), which reads the substantive content of each patent’s claims rather than relying on citation behavior.

The Kelly inventiveness score asks whether a patent’s text more closely resembles patents filed after it than patents filed before it. A patent whose text anticipates later filings is treated as inventive, while one that mostly recapitulates earlier filings is not.

The construction proceeds in four steps. First, each patent’s claims are tokenized with the Jieba segmenter, the standard tool for Chinese word segmentation. Second, the vocabulary is filtered: I keep tokens that appear in at least 10 patents, which yields 435,005 candidates, and then submit each candidate to the DeepSeek large language model, which classifies it as technical or non-technical vocabulary. The 199,782 tokens classified as technical form the final vocabulary. Without this filter, common characters and grammatical particles would dominate the token distribution without carrying technical signal. Third, each patent is represented as a term frequency–inverse document frequency (TF-IDF) vector over this vocabulary: the count of each token in the patent’s claims, weighted by the log of the inverse share of patents containing that token, so that rare technical vocabulary receives large weight and generic vocabulary little. The inverse-document-frequency weights are computed from patents filed in the ten years before the focal patent, so that the weights use only information available at the filing date.

Fourth, for each patent i filed in year t_i , I compute the average cosine similarity between its TF-IDF vector and those of patents filed in the ten years before t_i , and the corresponding average over patents filed in the ten years after t_i . Both averages are taken within the patent’s IPC section, so that

similarity reflects technological proximity rather than coincidental overlap across unrelated fields. The inventiveness score is the ratio of the two,

$$\text{inv}(i) = \frac{\text{sim}_{\text{fwd}}(i)}{\text{sim}_{\text{back}}(i)},$$

and a patent with $\text{inv}(i) > 1$ is more similar to subsequent filings than to prior ones. Relative to Kelly et al. (2021), three choices adapt the measure to Chinese patents: the Chinese-language tokenization, the filtering of vocabularies by a large language model, and the restriction of comparisons to patents in the same IPC section.

To convert the continuous score into a binary breakthrough indicator, I take the log of $\text{inv}(i)$ and subtract its year mean. The year demeaning removes a secular upward trend in raw inventiveness over the sample period, which arises mechanically as the corpus of comparison patents grows. I then flag patents in the top 10% of the demeaned distribution as breakthrough patents. This yields roughly 10% of filings in every year by construction.

Appendix F. Reduced-Form Robustness

This appendix tests the robustness of Table 1 along five dimensions: alternative cutoffs for the subsidy concentration measure, breakthrough patents as the dependent variable, two-way clustering, sample composition, and a pre-trend test of the Bartik instrument.

F.1 Alternative cutoffs

The subsidy concentration measure Conc_{jt}^p defines large firms as those above the 90th percentile of within-cell employment. Table F.1 reruns the column 3 specification at four alternative cutoffs: $p \in \{50, 75, 95, 99\}$. The small-firm coefficient is negative and significant at every cutoff under both estimators. At $p = 50$ and $p = 75$ the magnitude is comparable to the baseline estimate. At $p = 95$ and $p = 99$ both small- and large-firm coefficients grow in absolute value, reflecting the smaller and less stable group of firms at the extreme tail. The baseline $p = 90$ sits in the middle of the range.

F.2 Breakthrough patents

To check whether the crowding-out result is driven by strategic filing rather than genuine innovation, I replace the dependent variable with $\log(1 + \text{BT}_{jt,g})$, where breakthrough patents (BT) are the top decile of the Kelly inventiveness measure applied to Chinese patent text (Appendix E).

Table F.2 reruns the four-column specification with this dependent variable. The small-firm coefficient is negative and significant in OLS across all four columns, and remains negative under IV with magnitudes comparable to the baseline estimate, though with wider standard errors. The large-firm coefficient is small and statistically indistinguishable from zero in columns 2–4 under both estimators.

Table F.1: Alternative concentration cutoffs for the column 3 specification

	Concentration cutoff p				
	50	75	90	95	99
<i>Panel A: OLS</i>					
$\text{Conc}^p \times \text{Small}$	−0.889*** (0.319)	−0.738*** (0.172)	−0.682*** (0.171)	−0.713*** (0.196)	−0.951*** (0.284)
$\text{Conc}^p \times \text{Large}$	0.395 (0.319)	0.063 (0.213)	0.240 (0.189)	0.375** (0.183)	0.717*** (0.214)
<i>Panel B: Bartik IV</i>					
$\text{Conc}^p \times \text{Small}$	−8.394*** (2.889)	−3.856*** (1.179)	−6.703*** (1.475)	−11.101*** (2.255)	−25.445*** (5.428)
$\text{Conc}^p \times \text{Large}$	1.060 (1.862)	−0.045 (0.887)	1.217 (1.028)	3.865*** (1.379)	16.758*** (5.069)
First-stage F (S/L)	56 / 62	119 / 123	95 / 89	78 / 74	60 / 64
Observations (OLS / IV)	5,664 / 4,077	5,664 / 4,077	5,664 / 4,077	5,664 / 4,077	5,664 / 4,077

Notes: Each column reports the column 3 specification of Table 1 at a different concentration cutoff p . The dependent variable is $\log(1 + \text{Patents}_{jt,g})$. Controls match Table 1, column 3, with Conc^p and RevConc^p replaced at the corresponding percentile. Standard errors, clustered at the 3-digit industry level, are reported in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table F.2: Breakthrough patents as the dependent variable

	(1)	(2)	(3)	(4)
<i>Panel A: OLS</i>				
$\text{Conc}^{90} \times \text{Small}$	−2.021*** (0.119)	−0.542*** (0.137)	−0.546*** (0.137)	−0.371*** (0.075)
$\text{Conc}^{90} \times \text{Large}$	2.387*** (0.218)	0.258 (0.262)	0.265 (0.256)	0.123 (0.078)
<i>Panel B: Bartik IV</i>				
$\text{Conc}^{90} \times \text{Small}$	−2.214*** (0.170)	−2.142* (1.121)	−2.174* (1.130)	−1.340** (0.611)
$\text{Conc}^{90} \times \text{Large}$	2.264*** (0.183)	−1.105 (1.103)	−1.358 (1.144)	−0.618 (0.694)
First-stage F (S/L)	7,374 / 6,972	95 / 92	95 / 89	195 / 175
Observations (OLS / IV)	5,664 / 4,077	5,664 / 4,077	5,664 / 4,077	13,431 / 9,900

Notes: Same specification as Table 1, with the dependent variable replaced by $\log(1 + \text{BT}_{jt,g})$, where BT is the count of breakthrough patents (top-decile inventiveness, Appendix E). Standard errors, clustered at the 3-digit industry level, are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

F.3 Two-way clustering

The baseline specification clusters standard errors at the 3-digit industry level. If the residuals are also correlated across industries within a year, for instance because of common macroeconomic shocks to subsidy allocation, then single-industry clustering may understate the true sampling variance. Table F.3

reruns Table 1 with two-way clustering at industry and year.

The point estimates are identical to Table 1 by construction. Under two-way clustering, standard errors widen for the IV estimates in columns 2–4, but the small-firm coefficient stays significant at the 1% level.

Table F.3: Two-way clustering by industry and year

	(1)	(2)	(3)	(4)
<i>Panel A: OLS</i>				
$\text{Conc}^{90} \times \text{Small}$	−3.584*** (0.182)	−0.675*** (0.164)	−0.682*** (0.153)	−0.557*** (0.098)
$\text{Conc}^{90} \times \text{Large}$	2.802*** (0.121)	0.242 (0.219)	0.240 (0.208)	0.215* (0.101)
<i>Panel B: Bartik IV</i>				
$\text{Conc}^{90} \times \text{Small}$	−4.168*** (0.233)	−6.653*** (1.736)	−6.703*** (1.669)	−4.700*** (1.070)
$\text{Conc}^{90} \times \text{Large}$	3.350*** (0.187)	1.615 (0.951)	1.217 (0.954)	1.167 (0.888)

Notes: Same specification as Table 1, with standard errors using two-way clustering at the 3-digit industry and year level (Cameron et al., 2011). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

F.4 Sample splits

Table F.4 reruns the column 3 specification under four sample restrictions: dropping the top 1% of firms by employment, the top 1% by subsidy receipt, firms reporting subsidies in excess of revenue, and firms designated as technology SMEs. Each restriction addresses a potential concern about the baseline sample: that the result is driven by a few extreme firms, by misreporting, or by policy-favored firms in a separate program.

The small-firm coefficient is negative and significant across every split, with point estimates in the range of −0.65 to −1.19 for OLS and −5.53 to −7.33 for IV. The drop-top-1%-by-subsidy split, which removes the firms that receive the most subsidies, yields the smallest IV estimate in absolute value (−5.53). The other splits leave the estimate close to the baseline.

F.5 Pre-trend test

The Bartik instrument B_{jt} exploits persistent differences across provinces in their large-firm subsidy bias, weighted by industry-specific provincial employment shares from 2008. A natural concern is that these baseline shares may correlate with pre-existing trends in patenting that differ across size groups. I check this by regressing the within-cell change in patenting ($\Delta y_{jt,g} = y_{jt,g} - y_{j,t-2,g}$) on $B_{jt} \times \text{Small}$ and $B_{jt} \times \text{Large}$, following Jaeger et al. (2018).

The coefficient on the small-firm interaction is 0.091 with $p = 0.12$, statistically indistinguishable from zero. The small-firm crowding-out coefficient in Table 1 is therefore not driven by pre-existing

Table F.4: Sample splits for the column 3 specification

Sample	OLS		Bartik IV	
	Small	Large	Small	Large
Baseline (Tab. 1, col 3)	−0.682*** (0.171)	0.240 (0.189)	−6.703*** (1.475)	1.217 (1.028)
Drop top-1% emp.	−0.792*** (0.172)	0.262 (0.164)	−7.327*** (1.666)	2.303** (1.121)
Drop top-1% subsidy	−1.194*** (0.223)	−0.081 (0.222)	−5.533*** (1.095)	1.402 (0.865)
Drop subsidy > revenue	−0.655*** (0.172)	0.276 (0.199)	−6.760*** (1.505)	1.357 (1.090)
Drop tech-SME	−0.678*** (0.172)	0.128 (0.186)	−6.058*** (1.516)	1.111 (1.130)

Notes: Each row applies a firm-level filter to the working sample, then re-runs the column 3 specification of Table 1. Standard errors clustered at the 3-digit industry level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

differential patenting trends between industries the instrument predicts to have more or less subsidy concentration. The coefficient on the large-firm interaction is 0.155 ($p < 0.01$). Its magnitude is small relative to the second-stage IV estimate, though statistically significant. The large-firm IV estimate should be interpreted with this in mind.

Appendix G. Standard Errors

This appendix documents the construction of standard errors for the SMM estimates reported in Table 2.

Sandwich estimator. I compute standard errors for $\hat{\theta}$ using the SMM variance estimator of [Gouriéroux and Monfort \(1996\)](#), following [Garcia-Macia et al. \(2019\)](#):

$$\widehat{\text{Var}}(\hat{\theta}) = (G'WG)^{-1} G'W \hat{\Omega} WG (G'WG)^{-1}, \quad (\text{G.1})$$

where $G \in \mathbb{R}^{7 \times 6}$ is the Jacobian of model moments with respect to parameters at $\hat{\theta}$, $W \in \mathbb{R}^{7 \times 7}$ is the weighting matrix from (24), and $\hat{\Omega} \in \mathbb{R}^{7 \times 7}$ is the sampling covariance of the data moments. The model has no simulation step, so the simulation-noise term in the general [Gouriéroux and Monfort \(1996\)](#) formula is zero.

Bootstrap of $\hat{\Omega}$. I estimate the moment covariance with a firm-cluster bootstrap of the Chinese Tax Survey microdata. For each of $B = 500$ replicates, I resample $n = 52,091$ firms with replacement from the working sample and keep every firm-year observation from each sampled firm. If a firm is drawn twice, its full history appears twice in the bootstrap sample. I then recompute the inverse-probability weights that match the Chinese Economic Census firm size distribution, using the bin shares of the

bootstrap draw rather than the original sample, and rerun the full moment pipeline from Section 3. Firm-cluster resampling preserves the within-firm serial correlation that drives the slope moments. For each moment, $\hat{\Omega}$ uses the variance of the estimator that produces the corresponding data target. The bootstrap covariance of the seven moments forms $\hat{\Omega}$ and propagates the uncertainty in the data moments to the six estimated parameters through (G.1).

Numerical Jacobian. I compute the moment Jacobian G by central differences in *log-parameter space*:

$$(G_{\log})_{ik} = \left. \frac{\partial m_i}{\partial \log \hat{\theta}_k} \right|_{\hat{\theta}} \approx \frac{m_i(\hat{\theta} \cdot e^{+h} e_k) - m_i(\hat{\theta} \cdot e^{-h} e_k)}{2h}, \quad (\text{G.2})$$

with step size $h = 10^{-2}$, a one-percent relative perturbation of each parameter. Differentiation in log space applies a common relative step to parameters of different magnitudes, keeps every perturbed parameter strictly positive, and multiplies each column of the raw Jacobian by its parameter value ($G_{\log} = G_{\text{raw}} \cdot \text{diag}(\hat{\theta})$), which places the columns of G on a comparable scale and improves the conditioning of $G'WG$.

Reporting scale. Applying (G.1) with $G_{\log}$ yields $\widehat{\text{Var}}(\log \hat{\theta})$. Table 2 reports standard errors on the log scale, $\text{SE}(\log \hat{\theta}_k)$, which are unit-free and can be read as relative standard errors of the parameters. Confidence intervals are multiplicative, $[\hat{\theta}_k \exp(-1.96 \text{SE}(\log \hat{\theta}_k)), \hat{\theta}_k \exp(+1.96 \text{SE}(\log \hat{\theta}_k))]$. This form respects the strict positivity of the parameters, whereas the symmetric interval $\hat{\theta}_k \pm 1.96 \hat{\theta}_k \text{SE}(\log \hat{\theta}_k)$ has a negative lower bound whenever $\text{SE}(\log \hat{\theta}_k) > 1/1.96 \approx 0.51$. Raw-scale standard errors follow from the delta method,

$$\widehat{\text{Var}}(\hat{\theta}) = \text{diag}(\hat{\theta}) \widehat{\text{Var}}(\log \hat{\theta}) \text{diag}(\hat{\theta}),$$

so that $\text{SE}(\hat{\theta}_k) = \hat{\theta}_k \cdot \text{SE}(\log \hat{\theta}_k)$.

Appendix H. Supplementary Estimation Results

H.1 The Pat/R&D profile

Figure H.1 plots the size profile of Pat/R&D implied by the model as the process-efficiency cost scale c_ψ varies, holding the remaining parameters at their estimated values. At the estimated $\hat{c}_\psi = 0.32$, the profile falls from 0.24 at $n = 1$ to 0.04 at $n = 15$. As c_ψ rises, the profile flattens, and in the Klette–Kortum limit it is constant in n , as the analytical argument of Section 4.3 predicts.

H.2 Re-estimation without process efficiency

To shut off the process efficiency mechanism, I re-estimate the model under the restriction $c_\psi \rightarrow \infty$, minimizing the SMM objective over $(\theta, \lambda, H, \alpha_e)$. The fitted $\log(\text{Pat/R\&D})$ slope is +0.12 rather than −0.56, and the other three model-implied moments in Table H.1 are further away from their empirical

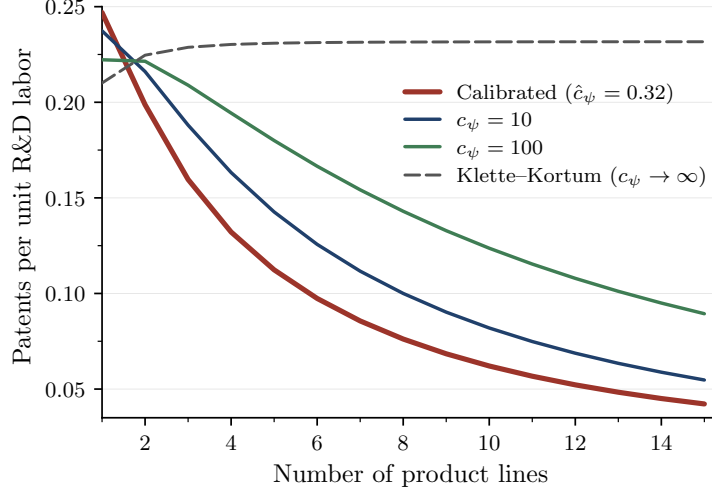


Figure H.1: Patents per unit of R&D labor as c_ψ varies

Notes: The figure plots the model-implied schedule $\text{Pat}(n)/\text{R\&D}(n)$ against the number of product lines n at four values of the process-efficiency cost scale c_ψ : 0.03, the estimated $\hat{c}_\psi = 0.32$, 30, and the Klette-Kortum limit $c_\psi \rightarrow \infty$. The remaining parameters ($\omega, \theta, \lambda, H, \alpha_e$) are held at their estimated values.

counterparts than under the full model.

Table H.1: Model fit: full versus restricted model

Moment	Data	Full model	Restricted
Aggregate growth	0.0222	0.0231	0.0407
IQR(log Emp)	1.39	1.08	0.60
log(R&D/Rev) slope	-0.61	-0.52	-1.24
log(Pat/R&D) slope	-0.56	-0.55	+0.12

Notes: The restricted model shuts down process efficiency by imposing $c_\psi \rightarrow \infty$ and re-estimates the remaining parameters against the benchmark SMM objective. The restricted estimates are $\hat{\theta} = 0.36$, $\hat{\lambda} = 0.55$, $\hat{H} = 1.00$, and $\hat{\alpha}_e = 2.78$. Slopes are coefficients from regressions of the indicated log ratio on log employment.

H.3 The crowding-out coefficient in the model

As a check on the model's general-equilibrium channels, I ask whether the calibrated model reproduces the reduced-form crowding-out coefficient from the preferred specification of Table 1. The exercise treats the model as the data-generating process for a controlled policy experiment. Holding all parameters at their estimates, I impose nineteen subsidy schedules that progressively raise the rate at the top of the size distribution while holding the small-firm rate fixed at its baseline level, and solve for the equilibrium of each. I then regress the gap in log per-firm patenting between small and medium firms, $\log(\text{Pat}_S^{\text{avg}}) - \log(\text{Pat}_M^{\text{avg}})$, on subsidy concentration Conc^{90} across the solved equilibria, treating the model as a single industry-year. Comparing the model's coefficient with the empirical estimate helps us understand whether the model's general-equilibrium channels are strong enough to account for the

crowding out measured in the data.⁵ The model-implied small-firm coefficient is -0.38 ($R^2 = 0.97$), slightly over half of the empirical column-3 OLS estimate (-0.68). The gap may be attributed to two structural features of the model: its compressed firm-size distribution, which dampens the wage and blocking responses to subsidy concentration, and the absence of channels outside the model, such as credit constraints that bind hardest on small firms, which may further suppress small-firm innovation in the data.

H.4 The calibrated economy

Figure H.2 plots two features of the economy at the parameter estimates of Table 3 of the main text. Panel (a) shows the share of R&D labor devoted to process efficiency rising from 13% at single-line firms to 67% at firms with eleven product lines. Panel (b) shows the probability that an attempted innovation against a size- n incumbent fails, which rises from near zero for single-line firms to over 0.12 for firms with eleven lines or more.

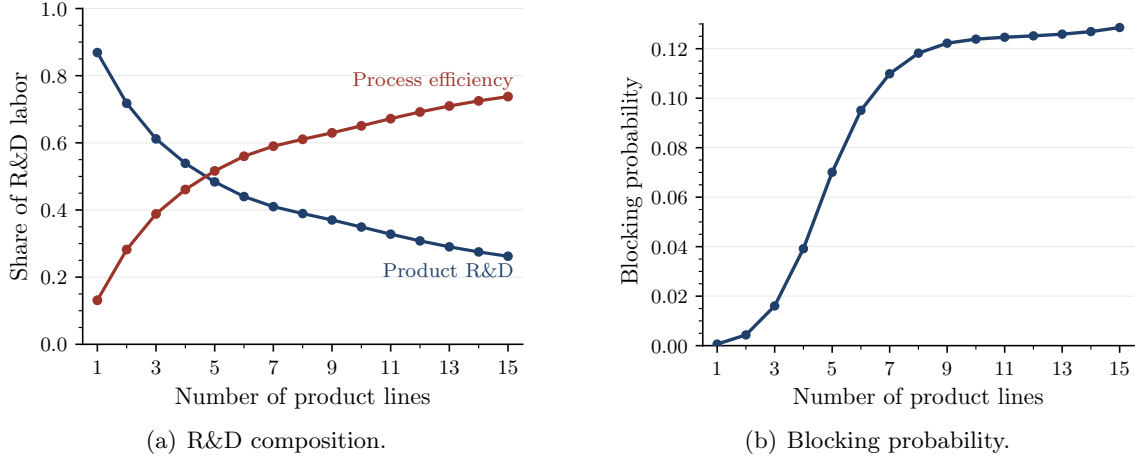


Figure H.2: Features of the calibrated economy

Notes: Panel (a) plots the composition of R&D labor against the number of product lines n . Panel (b) plots the probability that an attempted innovation against a size- n incumbent fails. Both panels are computed at the parameter estimates of Table 3 of the main text.

⁵I start from a baseline where all firms have a uniform 5% effective rate of discount on the cost of R&D labor, then incrementally increase the rate to firms in the p90–p99 and >p99 size bins by employment. Each incremental increase adds 2% to the discount for the >p99 bin and 1% for the p90–p99 bin (matching the approximate distribution in the data), holding the discount for the other firms unchanged. The nineteen subsidy schedules thus generate variation in Conc^{90} between 0.28 and 0.80. I then run a regression using these nineteen fixed points to assess the effect of Conc^{90} on small-firm patenting. Small firms are those below p47 by employment and medium firms are those between p47 and p86 by employment, which matches the data's 35- and 395-employee cutoffs. The dependent variable X^*B is the per-firm rate of successful product innovations (innovation effort times success probability), which is the model analog of patents per firm.

H.5 Consumption-equivalent variation

Welfare is evaluated on the balanced growth path using the household's CRRA preferences, with initial consumption net of the fiscal cost, $C_0 = \hat{Y} - T$. Let C_0, g_0 denote baseline consumption and growth and C_1, g_1 the alternative allocation, the CEV ξ solves $W(\xi C_0, g_0) = W(C_1, g_1)$ and equals

$$\xi = \frac{C_1}{C_0} \left[\frac{\rho - (1 - \vartheta) g_0}{\rho - (1 - \vartheta) g_1} \right]^{1/(1-\vartheta)}. \quad (\text{H.1})$$

I report $\xi - 1$ in percentage terms.

Appendix I. Computational Algorithm

The algorithm solves the model on the symmetric balanced growth path, where $Q = n$. It has two nested loops. The inner loop takes the skilled wage w_s as given and computes the BGP equilibrium objects (the balanced growth path defined in Section 2.4). The outer loop bisects on w_s to clear the skilled labor market.

The inner loop proceeds as follows:

1. Initialize w_s , the aggregate innovation rate $\bar{\tau}$, the active line share Φ_A , the average process efficiency $\bar{\Psi}$, and the firm size distribution $\mu(n)$ on a grid $n \in \{1, \dots, N_{\max}\}$.
2. Compute the normalized profit rate $\tilde{p}$ from equation (B.2).
3. Compute optimal process efficiency $\psi^*(n)$ from (5). Under a subsidy schedule, map firm sizes to employment, assign subsidy rates $\sigma_s(n)$, and use the effective wage $(1 - \sigma_s(n))w_s$ in place of w_s .
4. Compute the blocking probabilities $B(n)$ and per-line displacement rates $\tau(n)$ from equations (14) and (15).
5. Solve the Bellman equation (10) for $\hat{V}(n)$ by policy iteration, using the Thomas algorithm for the resulting tridiagonal system.
6. Obtain optimal innovation rates $X^*(n)$ from the first-order condition (11).
7. Compute the entry rate X_e from the free-entry condition (16), by bisection when $\alpha_e > 0$ and as residual labor when $\alpha_e = 0$.
8. Update the stationary distribution $\mu(n)$ by solving the flow equations (17), again as a tridiagonal system with the entry source at $n = 1$.
9. Update Φ_A , $\bar{\Psi}$, $\bar{\tau}$, and the firm mass $F = \Phi_A/\bar{n}$.
10. Check convergence. If μ has not converged, apply the damping update $\mu \leftarrow (1 - \delta)\mu + \delta\mu_{\text{new}}$ with $\delta = 0.4$ and return to step 2.

To save computational time, I set $N_{\max} = 80$ initially and raise it to $N_{\max} = 640$ only when more than 0.5% of the mass of μ ends up above $0.85N_{\max}$. I use a convergence tolerance of 10^{-7} on the inner loop and 10^{-6} on the outer bisection.

Appendix J. Robustness of the Quantitative Results

This appendix reports parameter estimates and welfare outcomes for the robustness exercises of Section 5.4.

J.1 Parameter estimates

Table J.1 reports the parameter estimates across the five specifications. All estimates remain within roughly one standard error of the benchmark, with the largest movements in the entry-cost parameters H and α_e , which partly substitute for each other along the free-entry condition.

Table J.1: Parameter estimates across robustness specifications

Parameter	Description	Benchmark	+ Yield	+ Process	$\gamma = 0.4$	$\gamma = 0.6$
ω	Process efficiency cost convexity	3.279	3.202	3.223	3.276	3.198
c_{ψ}	Process efficiency cost scale	0.323	0.260	0.339	0.361	0.398
θ	Incumbent innovation capacity	0.149	0.145	0.149	0.149	0.166
λ	Quality step size	1.079	1.177	1.072	1.047	1.005
α_e	Entry-cost convexity	3.000	2.615	3.000	2.914	3.000
H	Entry-cost level	0.527	0.328	0.534	0.476	0.598

Notes: Each column re-estimates the model under a different specification. “+ Yield” keeps the seven benchmark moments and adds an eighth targeted moment, the ratio of patents per subsidy yuan for subsidized firms in the p50–p90 size bin relative to firms below p50, computed within 4-digit industry $\times$ year cells. “+ Process” instead adds the share of large product innovators that also conduct process innovation (87.1%) as the eighth moment (Section 5.4). The γ columns fix the R&D production-function curvature and re-estimate the remaining six parameters against the benchmark moments. The externally calibrated parameters remain unchanged.

J.2 Welfare outcomes

Table J.2 reports the consumption-equivalent welfare gains under each specification for the three counterfactual policies of Section 5. The conclusions are unchanged across specifications: removing the existing subsidy regime yields a welfare gain of roughly 1% in consumption-equivalent terms, the budget-neutral optimum yields 10–14%, and the variable-budget optimum yields 18–22%.

The variable-budget gain varies the most across specifications, from 18.0% to 22.4%, because this counterfactual is the most sensitive to the estimated quality step λ and to the R&D production-function curvature γ .

Table J.2: Welfare gains across robustness specifications (CEV $\xi - 1$, %)

Counterfactual	Benchmark	+ Yield	+ Process	$\gamma = 0.4$	$\gamma = 0.6$
No subsidy	1.02	1.11	1.03	1.08	0.97
Budget-neutral optimum	10.73	12.33	10.69	13.64	12.59
Variable-budget optimum	18.20	21.54	18.04	21.39	22.40

Notes: Each entry is the consumption-equivalent welfare gain (CEV $\xi - 1$ in percent) relative to the baseline calibration of the corresponding specification, computed at $\rho = 0.020$ and $\vartheta = 2$. “+ Yield” adds the ratio of patents per subsidy yuan for subsidized firms in the p50–p90 size bin relative to firms below p50 as an eighth targeted moment, and “+ Process” adds the share of large product innovators that also conduct process innovation (Section 5.4). The γ columns fix the R&D production-function curvature and re-estimate the remaining parameters. The benchmark column reproduces Panel D of Table 5.

Appendix K. Firm Type Heterogeneity Extension

In this appendix I extend the baseline model with permanent firm-type heterogeneity in innovation productivity, following [Acemoglu et al. \(2018\)](#). Like the baseline dynamic analysis, the extension is solved on the symmetric balanced growth path, where $Q = n$. The extension shows that larger firms’ greater reliance on process R&D and the blocking externality operate within each type, so introducing types reshapes the cross-sectional distribution of firms across sizes without altering what firms do at any given size. The welfare gains reported in Section 5 are therefore lower bounds on the gains available to a planner who could also condition on type.

K.1 Setup

Each firm carries a permanent type $\theta_f \in \{\theta^h, \theta^l\}$ with $\theta^h > \theta^l$, drawn at entry success with $\Pr(\theta = \theta^h) = \alpha$. The innovation production function becomes

$$X = \theta_f h_X^{1-\gamma},$$

which replaces the type-uniform θ of equation (8). I leave the rest of the model unchanged: process-efficiency costs, the blocking condition, the entry technology, household preferences, and the final-good aggregator are all inherited from the baseline.

K.2 Process efficiency and blocking

The process-efficiency problem (4) does not involve θ , so the optimal choice

$$\psi^*(n) = \left(\frac{\tilde{p}n}{\omega w_s c_\psi} \right)^{1/(\omega-1)},$$

is a function of firm size alone. So is the profit function

$$\pi(n) = \tilde{p}n + \hat{D}n^\sigma, \quad \sigma = \frac{\omega}{\omega-1},$$

and the process R&D demand $h_\psi(n) = c_\psi[\psi^*(n)]^\omega$. Two firms of the same size but different types choose identical process efficiency, earn identical flow profit, and demand identical process R&D labor.

The blocking mechanism shares the same structure. The challenger–incumbent advantage $A_{n'',n} = (1 + \lambda)(1 + \psi^*(n'')) - (1 + \psi^*(n))$ depends on the attacker’s and the target’s sizes but not on their types. The success probability $B(n)$ and per-line displacement rate $\tau(n)$ keep the functional forms of equations (14) and (15), with the type-mixed attack intensity

$$\tilde{X}(n'') = F_h \mu_h(n'') X^h(n'') + F_l \mu_l(n'') X^l(n'')$$

replacing the single-type $\mu(n'')X(n'')$. Proposition 1 goes through because its proof only requires that $\psi^*(n)$ be increasing in n , and Proposition 2 goes through within each type.

K.3 Innovation and the firm-size distribution

I now turn to the dynamic optimization, which is where types come in. Let $\hat{V}_k(n)$ denote the quality-adjusted value of a type- k firm with n lines, with $\Delta V_k^+(n) = \hat{V}_k(n+1) - \hat{V}_k(n)$ and $\Delta V_k^-(n) = \hat{V}_k(n) - \hat{V}_k(n-1)$ as before. The Bellman equation generalizes (10) to

$$\hat{r} \hat{V}_k(n) = \pi(n) - w_s h_X^k(n) + X^k(n) B(n) \Delta V_k^+(n) - n \tau(n) \Delta V_k^-(n),$$

for $k \in \{h, l\}$, with the same effective discount rate $\hat{r} = \rho + (\vartheta - 1)g + \phi$ as in the baseline. The two Bellman equations share everything except the type-specific product R&D demand $h_X^k(n)$.

Differentiating with respect to X^k and using $h_X^k = (X^k/\theta_k)^{1/(1-\gamma)}$ gives the type- k first-order condition

$$\frac{w_s}{(1-\gamma)\theta_k} \left(\frac{X^k(n)}{\theta_k} \right)^{\gamma/(1-\gamma)} = B(n) \Delta V_k^+(n),$$

which solves to

$$X^k(n) = \theta_k^{1/\gamma} \left[\frac{(1-\gamma) B(n) \Delta V_k^+(n)}{w_s} \right]^{(1-\gamma)/\gamma}.$$

Because the exponent on θ_k is $1/\gamma > 1$, a small difference in type translates into a larger difference in the optimal innovation rate. A revealed-preference argument gives $\hat{V}_h(n) \geq \hat{V}_l(n)$ pointwise: the high type can mimic the low-type policy at strictly lower R&D wage cost.

Since types are permanent, the stationary distribution separates into two independent type-specific systems. Let $f_k(n)$ be the unnormalized type- k measure at size n . Its flow-balance equation takes the same form as the baseline (17),

$$[X^k(n)B(n) + n\tau(n) + \phi] f_k(n) = X^k(n-1)B(n-1)f_k(n-1) + (n+1)\tau(n+1)f_k(n+1) + \mathbf{1}\{n=1\}\alpha_k X_e,$$

where $\alpha_h = \alpha$ and $\alpha_l = 1 - \alpha$ partition the aggregate entry flow between the two types. The conditional pmf is $\mu_k(n) = f_k(n)/\sum_{n'} f_k(n')$, the type- k mass is F_k , and the marginal size distribution is $m(n) =$

$\eta \mu_h(n) + (1 - \eta) \mu_l(n)$ with $\eta = F_h/F$. High-type firms innovate faster through $X^h(n) > X^l(n)$, which means they grow faster and end up disproportionately at large n in the stationary distribution: $\mu_h(n)/\mu_l(n)$ rises in n .

K.4 Equilibrium

Types are revealed only after successful entry, so the free-entry condition (16) holds with $\hat{V}(1)$ replaced by the type-averaged value $\bar{V}(1) = \alpha \hat{V}_h(1) + (1 - \alpha) \hat{V}_l(1)$. Skilled-labor clearing sums product R&D over the two types, while process R&D aggregates through the marginal size distribution $m(n)$ alone, since $h_\psi(n)$ does not depend on type. The aggregate innovation rate $\bar{\tau}$ adds the type-specific incumbent flows to entry, and the balanced growth rate keeps the form of equation (19) with this $\bar{\tau}$ and the line shares $s(n) = n m(n)/\bar{n}$. Blocking depends only on size, so success and destruction probabilities aggregate cleanly across types. The welfare evaluation and the consumption-equivalent variation (H.1) carry over unchanged, with the $\bar{q}_0 = 1$ normalization applied as in the baseline.

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